

Johann Wolfgang Goethe-Universität Frankfurt am Main
Fachbereich Informatik und Mathematik

Bachelorarbeit

zur Erlangung des akademischen Grades
Bachelor of Science (B.Sc.) Informatik

An Ontological Perspective on LLM Beliefs

Submitted by: Qi An
Student ID: 8442657
First examiner: Prof. Dr. Visvanathan Ramesh
Themenausgabe: 27 June 2026
Submission date: 31 August 2026

Frankfurt am Main, 2026

Abstract

Large language models can answer a question and separately report how confident they are in that answer. These outputs are often treated as a single statement. Mechanistic work suggests a more specific process: information relevant to a later confidence report is represented near the generated answer and retrieved when the report is elicited. This thesis reproduces that protocol at reduced scale on Qwen2.5-7B-Instruct and extends its design from confidence to a family of verbal self-reports.

The reproduction separates answer generation from observation. Generation records the model’s answer and token statistics; observation reinserts the answer into a self-report prompt and reads one of ten ordered classes from the logits at the final cue. In the reduced experiment, confidence changes most clearly under intervention at the confidence cue. The answer-adjacent positions show little consistent separation, so this run does not reproduce the proposed temporal dissociation.

The extension, *vonto*, distinguishes the observation method from the independently computed ground truth. It studies confidence, commitment, nuance, and challenge across TriviaQA and lexical relations, and introduces task-specific measures for list elicitation and twenty questions. Calibration results cover all four datasets. The lexical-relation and TriviaQA profiles differ: the former is dominated by associations with binary correctness, whereas commitment and nuance in TriviaQA align most strongly with answer-sequence probability and answer entropy. List variety and twenty-questions impurity show modest positive associations with their designated measures, but list variety is similarly associated with the temperature control. Controlled steering results are limited to TriviaQA confidence and Synonyms commitment; the twenty-questions sample produced no variation in impurity reports and therefore could not supply the required high–low contrast.

The evidence supports a limited conclusion. The steering results show that the final report can be changed causally, but they do not distinguish cached retrieval from computation at the final cue. The narrow scope of the generalization evidence limits the broader ontological claim. “Belief” is used here as an operational family of prompted, measured self-reports, not as a claim that the model has human-like propositional attitudes.

Keywords: large language models; confidence; mechanistic interpretability; calibration; ontology; activation steering

Zusammenfassung

Große Sprachmodelle geben auf Fragen sowohl Antworten als auch verbalisierte Einschätzungen zu diesen Antworten aus. Diese Arbeit untersucht, ob eine solche Einschätzung erst beim Ausgeben berechnet wird oder ob relevante Information bereits während der Antwortgenerierung entsteht und später abgerufen wird. Dazu wird das Verfahren von Kumaran et al. (2026) in reduzierter Größe mit Qwen2.5-7B-Instruct reproduziert und mit dem Paket `vonto` auf weitere Selbstberichte erweitert.

In der reduzierten Reproduktion ändert sich die Konfidenz am deutlichsten unter Intervention an der Konfidenz-Cue-Position selbst; die antwortnahen Positionen zeigen keine konsistente Trennung. Der reduzierte Lauf reproduziert damit die vorgeschlagene zeitliche Trennung zwischen einer antwortnahen Repräsentation und dem späteren Konfidenz-Readout nicht. Die Erweiterung trennt Beobachtungsmethode und unabhängig berechnete Referenzgröße (Ground Truth) und untersucht deren Kalibrierung für TriviaQA, lexikalische Relationen, List Elicitation und Twenty Questions. Die aktuellen Aktivierungssteuerungsexperimente sind noch unvollständig; insbesondere ließ sich für Twenty Questions keine kontrastierende Testmenge bilden. Deshalb wird kein allgemeines Resultat über alle Konstrukte behauptet. Die Steuerungsergebnisse zeigen, dass sich der finale Bericht kausal verändern lässt; sie erlauben jedoch keine Unterscheidung zwischen einem gespeicherten Abruf und einer Berechnung erst am finalen Cue.

Note on the Use of AI Tools

AI-assisted tools were used for language revision and software development. I remain responsible for the research design, analysis, interpretation, and final text.

Bitte dieses Formular zusammen mit der Abschlussarbeit abgeben!

Erklärung zur Abschlussarbeit

gemäß § 35, Abs. 16 der Ordnung für den Bachelorstudiengang Informatik vom 17. Juni 2019:

Hiermit erkläre ich

(Nachname, Vorname)

Die vorliegende Arbeit habe ich selbstständig und ohne Benutzung anderer als der angegebenen Quellen und Hilfsmittel verfasst.

Ich bestätige außerdem, dass die vorliegende Arbeit nicht, auch nicht auszugsweise, für eine andere Prüfung oder Studienleistung verwendet wurde.

Zudem versichere ich, dass alle eingereichten schriftlichen gebundenen Versionen meiner vorliegenden Bachelorarbeit mit der digital eingereichten elektronischen Version meiner Bachelorarbeit übereinstimmen.

Frankfurt am Main, den

Unterschrift der/des Studierenden

Contents

Abstract	i
Zusammenfassung	ii
Note on the Use of AI Tools	iii
Declaration of Authorship	iv
1 Introduction	1
2 Background	3
2.1 Generation and observation phases	3
2.2 Positions and interventions	4
2.3 Calibration and ground truth	5
2.4 Ontological vocabulary	5
3 Related Work	7
3.1 Verbal confidence and uncertainty	7
3.2 Mechanistic interpretability	7
3.3 Datasets and epistemic constructs	8
3.4 Conceptual framing	8
4 Reproduction	9
4.1 Aim and setup	9
4.2 Interventions	9
4.3 Interpretation and scope	10
5 Method: Generalizing Beyond Confidence	11
5.1 Experimental layout	11
5.2 Discriminating logic	12

5.3	Operational scope	13
5.4	Observation methods, ground truths, and scope labels	13
5.5	Datasets	14
5.6	Calibration design	14
5.7	Steering design	15
6	Results	16
6.1	Scope of the analysis	16
6.2	Calibration	16
6.3	Steering results and generalization	22
7	Discussion: The Ontological Argument	26
8	Limitations	28
9	Conclusion and Future Work	29
	References	30

Chapter 1

Introduction

Large language models (LLMs) are routinely asked two things in one interaction. First, they answer a question. Second, they assess that answer by producing a number or a verbal confidence category. The second output is useful for selective prediction, abstention, and human review, but its status is unclear. A report such as “high confidence” could be a direct readout of a model state, a fluent continuation chosen because the prompt asks for it, or a property created by the measurement procedure itself.

Kumaran et al. (2026) treat the matter in a mechanistic way. Their intervention experiments distinguish the position where an answer ends from the final classification cue where a confidence class is read. They provide evidence that information predictive of the later report is present at a token adjacent to the answer. This information can be causally altered before the confidence report is elicited. If robust, this structure matters more than a calibration score alone: it offers an account of when information relevant to a verbal report becomes available and contradicts the idea that confidence is computed only at the final self-report readout.

This thesis first re-implements the protocol at smaller scale on Qwen2.5-7B-Instruct, then asks whether confidence is a special case. A mechanism confined to confidence may depend on the semantics of correctness; one shared by several kinds of self-report would support a broader account of prompted self-description. The title uses *beliefs* operationally for reports about a generated answer elicited by a specified prompt and readout procedure.

The extension features an observation-method $\times$ ground-truth framework. An observation method defines how a self-report is elicited and read during the observation phase (Phase 1). A ground truth is an independently computed scalar for the same trial. Separating the observation method from the ground truth prevents a self-report from being treated as valid merely because a convenient comparator is available. A confidence report can be compared with answer correctness; a variety report requires a property of the generated list; and an impurity report requires a score for the partition induced by a proposed question. Activation steering can be studied without a ground truth, whereas calibration requires both an observation method and a ground truth.

The thesis addresses four research questions. RQ1 concerns the status of an LLM belief

report: it may be understood as a representational state, a linguistic performance, or an observer-relative measurement artifact. These interpretations emphasize different aspects of the same phenomenon and need not be mutually exclusive. RQ2 asks whether the confidence dissociation can be reproduced at reduced scale on Qwen2.5-7B-Instruct. The expected pattern is an effect at the answer-adjacent position, before the confidence cue, with weaker effects at neighbouring control positions. RQ3 asks whether this pattern extends to other self-report constructs. This requires comparing intervention effects across positions, layers, and directions rather than merely obtaining a nonconstant response to a new prompt. RQ4 asks whether the combined evidence favours cache-and-retrieval over a just-in-time account of confidence, without attributing human-like belief to the model.

The empirical contribution has three parts. Chapter 4 presents the reduced reproduction and its methodological limits. Chapter 5 introduces *vonto*, which extends the protocol to commitment, nuance, challenge, variety, and impurity. Chapter 6 presents the calibration matrices and the two completed steering analyses.

The ontological analysis uses Benzon’s vocabulary as an interpretive framework rather than an empirical theory. The experiments are designed to distinguish local verbalization from information caching; the present evidence does not decide between them or determine whether an LLM holds beliefs in the philosophical sense.

The structure of the rest of the document is as follows: Chapter 2 sets out the elicitation protocol, the positions, the metrics, and the terminology. Chapter 3 places the study in the context of uncertainty estimation and mechanistic interpretability. Chapter 4 to Chapter 6 present the empirical work. Chapter 7 to Chapter 9 deal with the interpretation of the study’s scope, limitations, and implications.

Chapter 2

Background

2.1 Generation and observation phases

The experiment uses two forward passes. In the generation phase (Phase 0), the model receives an instruction and a question; the assistant prefill ends at Answer:. The resulting trial records the answer, its token log probabilities, and its next-token entropies. In the observation phase (Phase 1), the answer is inserted as ordinary text into a separate self-report prompt whose prefill ends at a cue such as Confidence:. The observation method assigns one of ten ordered classes from the first-token logits at that cue. Phase 1 does not inherit the residual stream from Phase 0. Information from generation can therefore reach the report only through the answer text or through representations formed when that text is processed in the observation prompt.

A formal experiment takes a trial to be a seed, an inquiry, a generated answer, the log probabilities of the answer tokens, and the entropies of the answer tokens:

$$\theta = (s, \mathcal{I}, a, \mathbf{g}, \mathbf{H}), \quad (1)$$

Here, $a \in \Sigma^*$ is the generated answer text, $\mathbf{g} = (g_1, \dots, g_{n_{\text{tok}}})$ contains its per-answer-token log probabilities, and $\mathbf{H} = (H_1, \dots, H_{n_{\text{tok}}})$ contains its per-step next-token entropies. An observation method ω has an ordered class list

$$C^\omega = (c_1^\omega, \dots, c_K^\omega), \quad K = |C^\omega| = 10, \quad (2)$$

and returns both a winning class and the corresponding class midpoint. The implemented observation methods and ground truths are

$$\Omega = \{\omega_{\text{conf}}, \omega_{\text{comm}}, \omega_{\text{nuan}}, \omega_{\text{chal}}, \omega_{\text{var}}, \omega_{\text{imp}}\}, \quad (3)$$

$$\Gamma = \{\gamma_{\text{corr}}, \gamma_{\text{prob}}, \gamma_{\text{ent}}, \gamma_{\text{chal}}, \gamma_{\text{lvar}}, \gamma_{\text{imp}}, \gamma_{\text{temp}}\}. \quad (4)$$

Equations (2)–(4) specify the observation method, its ten ordered report classes, and a ground truth evaluated on the same trial; each class is represented by its midpoint on the unit interval. It was verified that the Qwen tokenizer has pairwise distinct initial tokens for the class labels. Otherwise, an argmax over the first-token logits would cause the labels to be conflated.

2.2 Positions and interventions

There are five principal position labels. AC is the answer colon at the end of the Phase-0 prompt. PANL is the post-answer newline in the Phase-1 prompt. PANL+1 is the adjacent control token immediately after PANL. FCC is the format-class colon in the instruction block’s formatting example. CC is the final classification cue—historically called the confidence colon—where the class logits are read. The main comparison is between PANL and its PANL+1 and FCC controls. An effect at PANL without a corresponding control effect supports a specific answer-adjacent cache, whereas a flat curve at all positions shows that the intervention is ineffective.

Residual-stream steering involves adding a vector to the output of one decoder layer at a particular token position. For an observation method, the base vector is calculated as the difference between the mean residual activations of high-report and low-report extraction trials, then normalized to three per cent of the mean residual norm. The injected perturbation is α times this base vector; the retained generalized runs use $\alpha = 5$, and hence a perturbation norm equal to 15% of the mean residual norm. The extraction and test pools are disjoint, and both signs of the perturbation are used.

Let $z_{ij}^{(q)}$ be the class- j logit for test trial i under condition $q \in \{0, +, -\}$, where $q = 0$ is clean and $q = +$ or $q = -$ denotes the two steering directions. Let $y_i = \arg \max_j z_{ij}^{(0)}$ be the clean winning class, and let $m_i^{(q)}$ be the midpoint of the winning class under condition q . The first two readouts are computed per trial and the third is their aggregate class-change rate:

$$\begin{aligned} \Delta m_i^{(q)} &= m_i^{(q)} - m_i^{(0)}, \\ d_i^{(q)} &= \left(z_{iy_i}^{(q)} - \frac{1}{K-1} \sum_{j \neq y_i} z_{ij}^{(q)} \right) - \left(z_{iy_i}^{(0)} - \frac{1}{K-1} \sum_{j \neq y_i} z_{ij}^{(0)} \right), \\ r^{(q)} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_j z_{ij}^{(q)} \neq y_i \right]. \end{aligned} \tag{5}$$

Thus, $\Delta m_i^{(q)}$ is the self-report midpoint change, $d_i^{(q)}$ is the change in the clean winner’s logit relative to the mean of all other class logits while holding y_i fixed, and $r^{(q)}$ is the first-token change rate. For a sample $A = (a_1, \dots, a_N)$, the plotted layer curves use

$$\bar{A} = \frac{1}{N} \sum_{i=1}^N a_i, \quad \text{sem}(A) = \begin{cases} \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N (a_i - \bar{A})^2}, & N > 1, \\ \text{NaN}, & N \leq 1. \end{cases} \tag{6}$$

AC requires a different treatment, because it belongs to the generation phase. Steering AC cannot directly change an observation-phase read. The answer must be regenerated under intervention and then assessed with an unsteered observation-phase prompt. This regeneration control lies outside the reported steering sweep.

2.3 Calibration and ground truth

Calibration is assessed using Spearman rank correlation because the reports are ordinal and the ground-truth values have different scales. For paired samples $(u_i, w_i)_{i=1}^N$, let $\text{rk}(\cdot)$ denote the midrank transform. Spearman’s rho is the Pearson correlation of these midranks:

$$\rho(u, w) = \frac{\sum_{i=1}^N (\text{rk}(u_i) - \overline{\text{rk}(u)}) (\text{rk}(w_i) - \overline{\text{rk}(w)})}{\sqrt{\sum_{i=1}^N (\text{rk}(u_i) - \overline{\text{rk}(u)})^2} \sqrt{\sum_{i=1}^N (\text{rk}(w_i) - \overline{\text{rk}(w)})^2}}. \quad (7)$$

The value is undefined if either series is constant; such cells are shown as n/a. Paired percentile-bootstrap intervals accompany the principal point estimates because rank correlations need not have a normal sampling distribution (Haukoos & Lewis, 2005).

vonto maintains the observation-method report and its ground-truth counterpart as separate quantities. Confidence is paired with correctness. Correctness is determined by normalized containment of the generated answer and the accepted answer aliases; this rule is stricter than semantic grading and can reject valid paraphrases. Commitment is paired with the probability of the exact generated answer sequence. Nuance is paired with answer entropy. The Challenge report measures resistance, whereas its implemented ground truth measures susceptibility: a larger baseline-to-challenged defend-margin drop means greater caving, so the two scales have opposite polarity and their expected association is negative. Genuine and counterfeit challenge evidence are generated under the same sampling settings and differ in the answer they support. Variety is paired with list variance. Impurity is paired with the Gini impurity of the yes/no split induced by a Twenty Questions query. Temperature is a negative control stored in the List Elicitation seed but never presented directly to the model, so a correlation may indicate a behavioural pathway or leakage rather than calibration to a substantive target.

2.4 Ontological vocabulary

The thesis considers three interpretations of LLM belief. On a representational-state reading, a report provides evidence of an internal state that persists across prompt positions. On a linguistic-performance reading, the report is an utterance conditioned by the prompt and need not express a persistent state. On an observer-relative reading, the measured construct depends partly on the prompt, tokenizer, class scale, and extraction rule. The interpretations may apply simultaneously at different levels of description.

Benzon’s distinction among inheritance structures, paradigmatic transitions, and onto-

logical transitions informed the dataset design (Benzon, 1987). The synonym dataset assigns a positive correctness value only to interchangeable pairs; co-hyponyms and broader lexical relations serve as negative examples. The pairs come from human-annotated benchmarks that classify semantic relations rather than lexical similarity alone (Santus et al., 2016). In the twenty-questions task, each proposed question partitions the candidate universe, following Benzon’s use of the game to examine conceptual structure in ChatGPT (Benzon, 2023b). This vocabulary guides the analysis but is not treated as an empirical model of transformers.

Chapter 3

Related Work

3.1 Verbal confidence and uncertainty

Verbal confidence reports are commonly used to estimate uncertainty in large language models. Kadavath et al. (2022) found that sufficiently large models can assess whether an answer is true and estimate the probability that they know an answer. Comparing verbal, consistency-based, and hybrid elicitation methods, Xiong et al. (2023) found substantial overconfidence in verbal reports. These studies establish the practical value and limitations of self-evaluation but do not identify its mechanism. Kumaran et al. (2026) address that question with causal interventions, position-specific analyses, and linear probes. They propose that confidence-related information is formed near the answer, retained at an adjacent position, and retrieved when the confidence report is produced.

Some uncertainty estimators depend on the model’s behaviour rather than having it directly report its uncertainty. Semantic entropy combines answers that have the same meaning and then uses their distribution to identify confabulations (Farquhar et al., 2024). Such methods are useful baselines because answer-sequence probability may be a poor substitute for uncertainty at the answer level. The current study shares this concern. Its answer-entropy and answer-sequence-probability ground truths test whether observation-method reports covary with the specified computed properties, while activation steering investigates where information causally relevant to a report is available.

3.2 Mechanistic interpretability

Mechanistic interpretability relates model behaviour to computations at specific layers and token positions. The anchor study uses activation steering, patching, noising, representation swaps, probing, and attention blocking. These methods support different inferences: a probe shows that information is decodable, whereas a targeted intervention tests whether changing a representation changes a later readout. Correlation or probe performance alone cannot establish that a position causally contributes to a self-report.

This thesis tests whether a controlled intervention next to the answer has a different

downstream effect from interventions at neighbouring positions. It does not equate a localized vector with a complete semantic representation of confidence, since any model state may encode several correlated features.

3.3 Datasets and epistemic constructs

TriviaQA includes question-answer pairs together with accepted aliases and supports a correctness ground truth (Joshi et al., 2017). The lexical-relation data used by *vonto* distinguishes interchangeable synonyms, co-hyponyms, and broader relationships. The implemented pools draw on the EVALution and CogALex-V benchmark series, whose purpose is to classify lexical semantic relations (Santus et al., 2016). List Elicitation and Twenty Questions provide settings in which the generated answer is not evaluated by a single correct/incorrect distinction. They instead support the task-specific Variety and Impurity observation methods, paired with list variance and partition Gini impurity respectively.

Instead, the extension uses epistemic rather than basic affective constructs. Research into epistemic emotions separates knowledge-related states and metacognitive feelings from general affect, although it also cautions that the boundaries between them are disputed (Nerantzaki et al., 2021). The project does not draw any conclusion about a human emotional state from the model’s report; rather, the terms refer to prompt-defined observation methods.

3.4 Conceptual framing

Philosophical accounts of belief help define the object of inquiry but need not provide an empirical model of transformers. The inheritance-tree and transition terminology used here comes from Benzon (1987). Related work applies these distinctions to ChatGPT and Twenty Questions (Benzon, 2023b), proposes nested probability distributions as an intermediate descriptive level between text and implementation (Benzon, 2023a), and examines story prototypes that recur under changes in prompting (Benzon, 2024). These sources supply interpretive vocabulary rather than experimental evidence. The empirical claims rest on the trials and interventions; the philosophical analysis asks which account best fits those results, including the possibility that the extraction procedure partly constitutes the measured object.

Chapter 4

Reproduction

4.1 Aim and setup

The protocol of Kumaran et al. (2026) is reproduced in this chapter. The reduced profile uses Qwen2.5-7B-Instruct in BF16 evaluation mode, with greedy decoding and the no-context TriviaQA validation split. Qwen2.5 is an instruction-tuned version of the Qwen2.5 open-weight series (Qwen Team, 2024). The recorded behavioural run consists of 1,000 trials: 300 activation-pool trials, 40 calibration-pool trials, and a held-out remainder. This reduction maintains the prompts, positions, interventions, and metrics at a smaller sample scale. It therefore cannot be considered a numerical replication of the paper’s full-profile results.

The reported run uses exact matching against the accepted TriviaQA aliases rather than the GPT-4o-mini grader used in the anchor study. This stricter rule can classify semantically acceptable variants as incorrect and may therefore lower measured accuracy. A chat template with an assistant prefill places AC and CC at the final prompt tokens. Trials are omitted from the position-specific analysis when the tokenizer merges the post-answer newline with answer punctuation, since PANL would then no longer denote the intended position.

4.2 Interventions

The reported reproduction comprises the behavioural baseline and confidence steering described in Table 4.1. These components test calibration and position-controlled causal effects, respectively.

Table 4.1: Evidence in the reduced reproduction.

Evidence group	Model and prompt	Sample composition	What it supports
Behavioral baseline	Qwen2.5-7B-Instruct, categorical TriviaQA	1,000 generated trials	Calibration and behavioural checks
Confidence steering	Qwen2.5-7B-Instruct, categorical TriviaQA	200 prepared trials	Position- and layer-controlled causal read-out

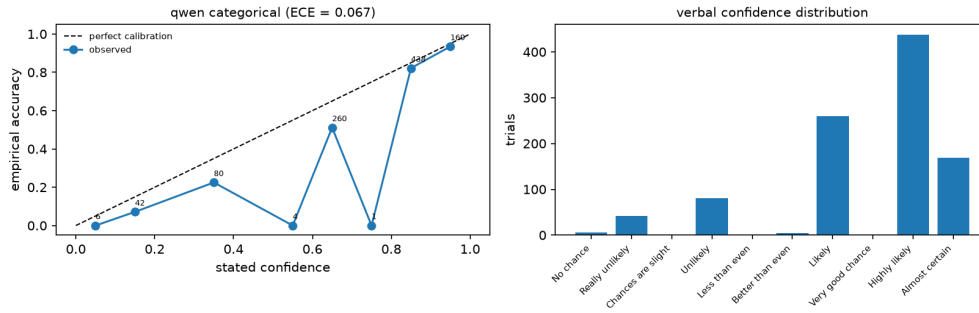


Figure 4.1: Calibration plot from the reduced categorical Qwen TriviaQA run.

In the reduced categorical run, Fig. 4.1 shows the confidence observation-method report plotted against bins of empirical correctness. The horizontal axis lists confidence class mid-points or bins, while the vertical axis displays observed correctness. The error bars describe sampling uncertainty in the bin statistic. The plot establishes that the elicitation and grading procedure produces a usable confidence readout; mechanistic interpretation requires the intervention results.

4.3 Interpretation and scope

The confidence steering result in Fig. 6.8 shows the clearest change at the confidence cue, while PANL and the neighbouring positions show little consistent separation. The reduced run therefore does not reproduce the answer-adjacent pattern and does not independently support cached retrieval. Direct numerical comparison with the anchor study is inappropriate because the model, layer sweep, sample size, and correctness rule differ. The standard-error bars describe variation within the reduced held-out sample and provide no estimate of variation across models.

The reproduction is restricted to Qwen2.5-7B-Instruct and the reported TriviaQA conditions. Its findings therefore do not establish generality across architectures, datasets, or reasoning models.

Chapter 5

Method: Generalizing Beyond Confidence

5.1 Experimental layout

The experiment has two linked analyses built from the same two-phase trial structure. Phase 0 generates the answer and Phase 1 elicits the observation; there is no Phase 2. The calibration analysis asks whether an elicited report covaries with an independently computed ground truth. The steering analysis asks whether the report can be causally changed at a particular layer and prompt position. Calibration is therefore a measurement-validity test, whereas steering is a mechanistic intervention test; success in either analysis does not imply success in the other.

The end-to-end procedure is organized as follows.

1. **Prepare a seed and inquiry.** A dataset record supplies the question or task, its task-specific metadata, and, where applicable, an accepted answer or candidate universe.
2. **Generate the answer (Phase 0).** The model produces an answer from the inquiry. The resulting trial stores the answer text together with its token log probabilities and next-token entropies.
3. **Elicit a report (Phase 1).** The generated answer is inserted as ordinary text into a separate observation prompt. At the final classification cue, the observation method reads the first-token logits of ten ordered class labels and assigns the midpoint of the winning class.
4. **Compute ground truths.** Ground-truth functions operate on the same saved trial, its seed, or an additional task-specific model call. They do not use the winning self-report class as their input.
5. **Evaluate calibration.** On one common trial pool per dataset, every applicable observation-method report is paired with every applicable ground-truth value. Spearman correlations form the calibration matrix; matched pairs test the intended construct,

off-diagonal pairs test alternative explanations, and hidden temperature acts as a negative control.

6. **Construct the intervention.** For each observation method, disjoint extraction trials are divided into relatively low- and high-report groups. At each layer, their mean residual activations are subtracted to form a base steering direction normalized to 3% of the mean residual norm. The intervention injects $\pm\alpha$ times that base vector.
7. **Test causal effects.** Both signs of the direction are injected into a separate balanced test set at PANL, PANL+1, FCC, or CC. Changes in report midpoint, clean-class logit difference, and winning first token are compared with the unsteered condition across layers. PANL is the answer-adjacent target, PANL+1 and FCC are positional controls, and CC verifies that the intervention can affect the final readout directly.

Table 5.1: Layout of the generalized experiment.

Stage	Data unit	Primary output	Role in the argument
Phase 0: generation	Seed and inquiry	Answer, token log probabilities, and token entropies	Creates the trial and answer-related quantities
Phase 1: observation	Saved Phase-0 trial	Ordered report class and midpoint	Measures the prompted self-report independently of the generation pass
Ground-truth evaluation	Same trial and task metadata	Scalar comparator	Defines the external relation tested by calibration
Calibration	Common dataset pool	Observation-method $\times$ ground-truth Spearman matrix	Tests convergent, discriminant, and negative-control relationships
Direction extraction	Low/high extraction subsets	Layer-specific normalized steering vector	Identifies a candidate representation of the report
Steering test	Disjoint balanced test subset	Layer-by-position causal-effect curves	Tests directionality, position specificity, and final-readout sensitivity

The reproduction in Chapter 4 uses this conceptual layout for confidence on TriviaQA at reduced scale, but it uses a different partition scheme from the generalized experiment. The generalized experiment retains the same Phase-0/Phase-1 separation and intervention logic while replacing the single confidence–correctness pair with the observation methods, ground truths, datasets, and controls specified below. Chapter 6 reports the conditions for which calibration or steering evidence is available.

5.2 Discriminating logic

Either outcome of the generalization test is informative. A shared answer-adjacent pattern across constructs would count against treating confidence as a special-purpose introspective state. Failure to generalize would instead suggest that confidence has a distinctive relation to correctness, the prompt, or learned response conventions. A null result is interpretable only

when the intervention changes the report at CC and the adjacent control positions have also been tested.

The framework separates mechanism from measurement validity. The mechanistic analysis asks whether an intervention can change a forthcoming report, regardless of whether that report corresponds to an external fact. The validity analysis asks whether the report ranks trials according to a stated, independently computed ground truth. Keeping these questions separate prevents a confidence result from being generalized merely because factual questions provide a convenient correctness criterion.

5.3 Operational scope

The analysis concerns a prompted and tokenized report, not an unobserved mental state. Its value depends on the wording, class labels, tokenizer, forced cue, and argmax rule. This observer dependence is both a limitation and part of the research question. Checks for unique class tokens and isolated newline positions make the dependence explicit.

Steering vectors are estimated from one trial pool and evaluated on another, reducing the risk that a vector captures idiosyncrasies of the test questions. By contrast, every cell in a calibration matrix uses the same trial pool so that diagonal and off-diagonal correlations are comparable. These correlations remain observational rather than causal.

5.4 Observation methods, ground truths, and scope labels

vonto generalizes the reproduction while preserving the original protocol; its key abstraction is a pair of independent objects evaluated on the same trial. The observation method determines how a self-report is elicited and read, while the ground truth independently computes a scalar score for the trial. Calibration is the cross-product analysis that correlates each observation-method report with each ground-truth value over a common trial pool. The trial is therefore generated in Phase 0, observed in Phase 1, and analysed across the pool during calibration.

The cross-dataset observation methods are confidence, commitment, nuance, and challenge, while their ground truths are correctness, answer-sequence probability, answer entropy, and challenge susceptibility. This grouping defines the shared analysis and does not imply that a construct is task-independent or psychologically universal. The analysis pairs are confidence with correctness, commitment with answer-sequence probability, nuance with answer entropy, and reported challenge resistance with challenge susceptibility. The last pair has opposite polarity by construction, so evidence of alignment appears as a negative rather than a positive rank correlation.

Other elements are specific to the task. The variety method is the list-elicitation observation method, with list variance as its main ground truth and temperature serving as a hidden-seed negative-control ground truth. Impurity refers to the twenty-questions observation method and is associated with the Gini impurity of the induced yes/no candidate

partition.

All observation methods employ ten ordered report classes. For confidence, the exact wording from the anchor paper is retained; all other methods use a common ten-step intensity scale whose class-initial Qwen tokens were checked for pairwise uniqueness. The reported value is the midpoint of the winning class. This standardisation enables comparisons between observation methods, but it does not demonstrate that the linguistic labels correspond to psychologically equally spaced quantities.

5.5 Datasets

TriviaQA provides factual questions together with accepted answer aliases. The lexical-relation dataset has three relation tiers: twins are interchangeable synonyms, siblings are co-hyponyms, and relatives represent broader lexical relationships. Its inquiry requires a Yes or No answer. List Elicitation requests a list associated with a selected keyword and varies the sampling temperature across seeds for each keyword. Unlike the other generation tasks, it does not stop at the first blank line, because a multi-line list is the intended answer. Twenty Questions consists of played-out games over a candidate-word universe; the turn history is created during dataset preparation because it is needed to generate each subsequent question. Seven category families are concrete and three are abstract, enabling a further subgroup analysis where the data support it.

Each calibration analysis uses 200 trials. The prepared lexical pool contains 201 pairs (67 in each relation tier), from which 200 are used for calibration; the list and twenty-questions datasets each contribute 200 trials. These sample sizes reflect computational constraints rather than a prospective precision calculation. Bootstrap intervals are therefore reported for the principal matched correlations, while the remaining heatmap cells are interpreted descriptively.

5.6 Calibration design

For each dataset, every observation-method $\times$ ground-truth cell is Spearman’s rho on a common trial set. The designated pairs test whether a report covaries with its chosen comparator; for challenge, resistance is compared with oppositely oriented susceptibility. Correctness uses normalized alias matching: case, English articles, punctuation, and repeated whitespace are removed, after which equality or containment in either direction counts as a match against any accepted alias. This procedure is deterministic but can reject semantically correct paraphrases. Genuine and counterfeit challenge evidence use the same generation procedure and sampling temperature, so their intended difference is the answer they support rather than their style. Off-diagonal cells test whether a report instead follows a generic property such as answer quality or length. Correlation is undefined for a constant vector and is reported as such rather than set to zero.

The temperature control is particularly important in this matrix. The sampling temper-

ature is selected and stored in the seed before Phase-0 generation and is not included in the Phase-1 self-report prompt. A non-zero relationship between report and temperature would indicate leakage or a behavioural pathway through the generated answer, not that the model directly reports the hidden seed value.

5.7 Steering design

For each construct, steering vectors are derived from high- and low-report extraction trials, with an optional restriction using the construct’s paired ground truth. Each completed analysis uses 30 extraction trials and the ten highest and ten lowest reports after this restriction. A disjoint pool of 60 candidate trials supplies a balanced 30-trial test set containing the 15 lowest and 15 highest clean reports. The base vectors are normalized to 3% of the mean residual norm, after which $+\alpha$ or $-\alpha$ times the vector is injected at PANL, PANL+1, FCC, or CC. The completed analyses use $\alpha = 5$, corresponding to an injected perturbation norm of 15% of the mean residual norm, and evaluate layers 0, 2, 4, $\dots$, 26, and 27. Measurements are taken in both directions on the balanced test set. The CC curve tests whether the intervention can affect the final classification readout; PANL+1 and FCC serve as positional controls. AC requires regenerating the answer under intervention in Phase 0 before an unsteered Phase-1 observation and is therefore treated as a separate control.

vonto uses relative rank splits to form nonempty, disjoint low- and high-report subsets. This makes the balanced-test sampling rule part of the measurement design. When the report is constant, no steering contrast can be formed.

Chapter 6

Results

6.1 Scope of the analysis

The analysis combines the 200-trial calibration samples with controlled steering results for TriviaQA confidence and Synonyms commitment. Calibration is evaluated through the observation-method $\times$ ground-truth matrices and their principal bootstrap intervals. The constant impurity sample is reported because it prevents the corresponding causal contrast.

6.2 Calibration

Table 6.1: Calibration datasets and measures.

Experiment	Dataset	Observation methods	Ground-truth columns	Trials
Baseline	Synonyms	confidence, commitment, nuance	correctness, answer-sequence probability, answer entropy	200
Baseline	TriviaQA	confidence, commitment, nuance	correctness, answer-sequence probability, answer entropy	200
Full matrix	Synonyms and TriviaQA	baseline methods plus challenge	baseline columns plus challenge	200 per dataset
List elicitation	20 temperatures across 10 keywords	variety	list variance; hidden temperature control	200
Twenty questions	20 games across 10 categories	impurity	Gini impurity	200

Table 6.1 summarises the calibration datasets used in the analysis.

The matched correlations in Table 6.2 use Spearman’s statistic as defined in Equation (7). The 95% paired percentile-bootstrap intervals use 2,000 resamples with a fixed random seed and are not adjusted for multiple comparisons (Haukoos & Lewis, 2005). All values were calculated from the trial-level data.

Within the stated samples, the designated confidence, commitment, nuance, variety, and impurity comparisons have associations in the expected direction. Challenge requires the opposite sign: the report increases with resistance, while the ground truth increases with

Table 6.2: Selected observation-method $\times$ ground-truth Spearman correlations, with 95% bootstrap intervals.

Dataset	Matched comparison	Spearman rho	95% interval	n
Synonyms	confidence $\times$ correctness	0.516	[0.416, 0.602]	200
Synonyms	commitment $\times$ answer-sequence probability	0.530	[0.419, 0.635]	200
Synonyms	nuance $\times$ answer entropy	0.489	[0.380, 0.588]	200
Synonyms	challenge resistance $\times$ challenge susceptibility	0.360	[0.233, 0.473]	200
TriviaQA	confidence $\times$ correctness	0.262	[0.128, 0.394]	200
TriviaQA	commitment $\times$ answer-sequence probability	0.405	[0.271, 0.517]	200
TriviaQA	nuance $\times$ answer entropy	0.627	[0.525, 0.710]	200
TriviaQA	challenge resistance $\times$ challenge susceptibility	-0.162	[-0.294, -0.009]	200
List elicitation	variety $\times$ list variance	0.224	[0.083, 0.365]	200
List elicitation	variety $\times$ temperature control	0.244	[0.109, 0.370]	200
List elicitation	temperature $\times$ list variance	0.183	[0.034, 0.317]	200
Twenty questions	impurity $\times$ Gini impurity	0.306	[0.171, 0.437]	200

susceptibility to contrary evidence. The Synonyms estimate is instead positive, contrary to the intended alignment. The TriviaQA estimate is negative with an interval excluding zero, as expected from the opposing scales, but its small magnitude gives limited evidence of alignment. The temperature associations are also positive, so the hidden control is not inert in this sample.

Figure 6.1 shows the three-by-three lexical-relation calibration matrix. Rows are observation methods, columns are ground truths, and each cell gives Spearman’s rho over the common trial pool. The designated comparisons are 0.516 for confidence with correctness, 0.530 for commitment with answer-sequence probability, and 0.489 for nuance with answer entropy; bootstrap intervals appear in Table 6.2. Commitment is more strongly associated with correctness (0.687) than with its designated probability ground truth, while nuance is more strongly associated with correctness in the opposite direction (-0.769) than with entropy. The off-diagonal correlations therefore preclude a claim of construct specificity.

Figure 6.2 uses the same layout for TriviaQA. The matched confidence $\times$ correctness correlation is 0.262, with a 95% bootstrap interval of [0.128, 0.394]. Commitment $\times$ answer-sequence probability is 0.405 [0.271, 0.517], and nuance $\times$ answer entropy is 0.627 [0.525, 0.710]. Unlike the lexical-relation matrix, probability and entropy are the strongest respective associations for commitment and nuance. The correctness column uses the recorded alias-matching ground truth. These comparisons apply only to that rule and trial pool; they do not establish calibration under a more permissive semantic judgment.

Figure 6.3 adds challenge resistance and challenge susceptibility to the lexical-relation matrix; the remaining axes match Figs. 6.1 and 6.2. Because the challenge scales have opposite polarity, alignment would produce a negative association. The observed value is instead 0.360, with a 95% interval of [0.233, 0.473]. Its positive sign is incompatible with the intended interpretation of the two scales and does not validate the challenge observation

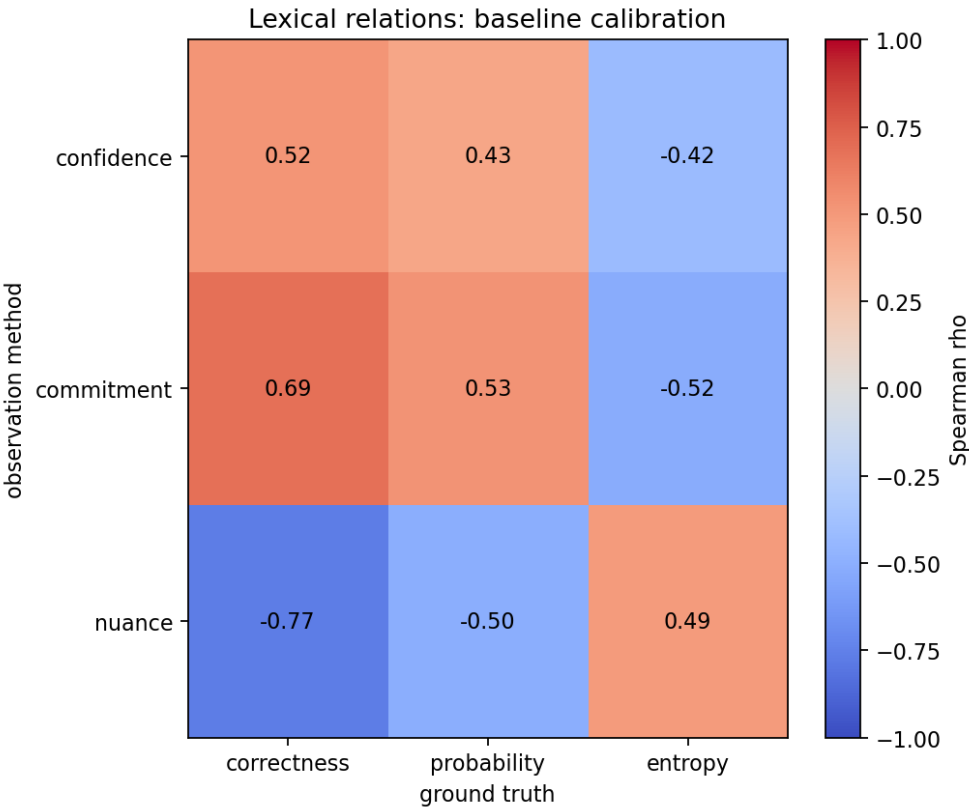


Figure 6.1: Baseline calibration matrix for lexical relations.

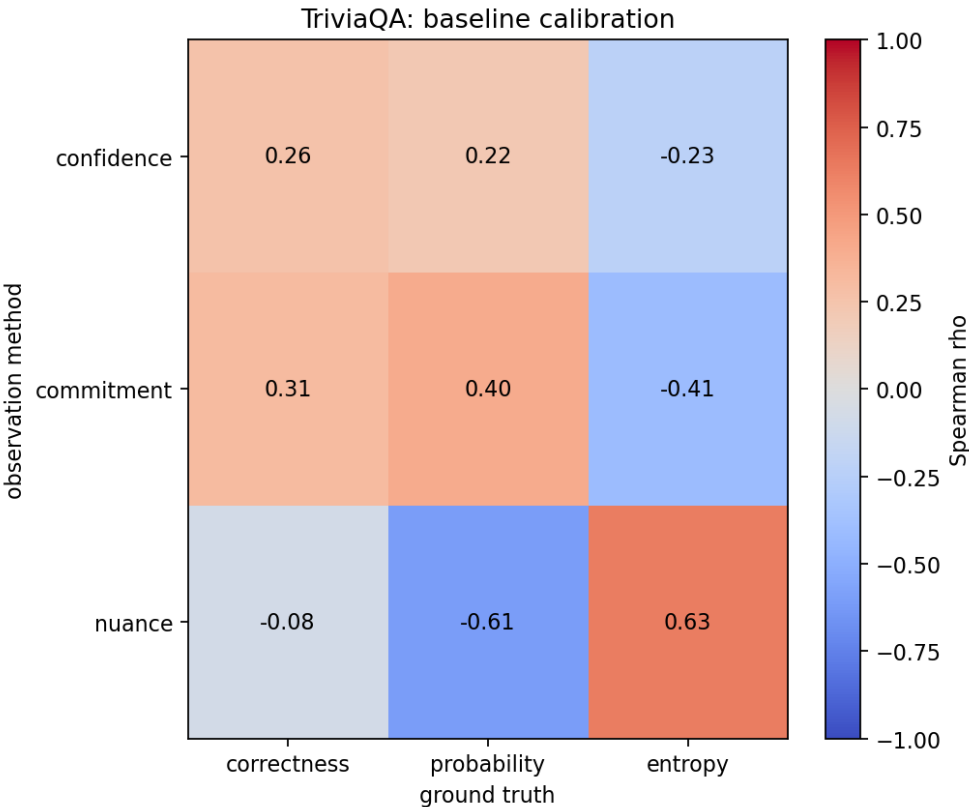


Figure 6.2: Baseline calibration matrix for TriviaQA.

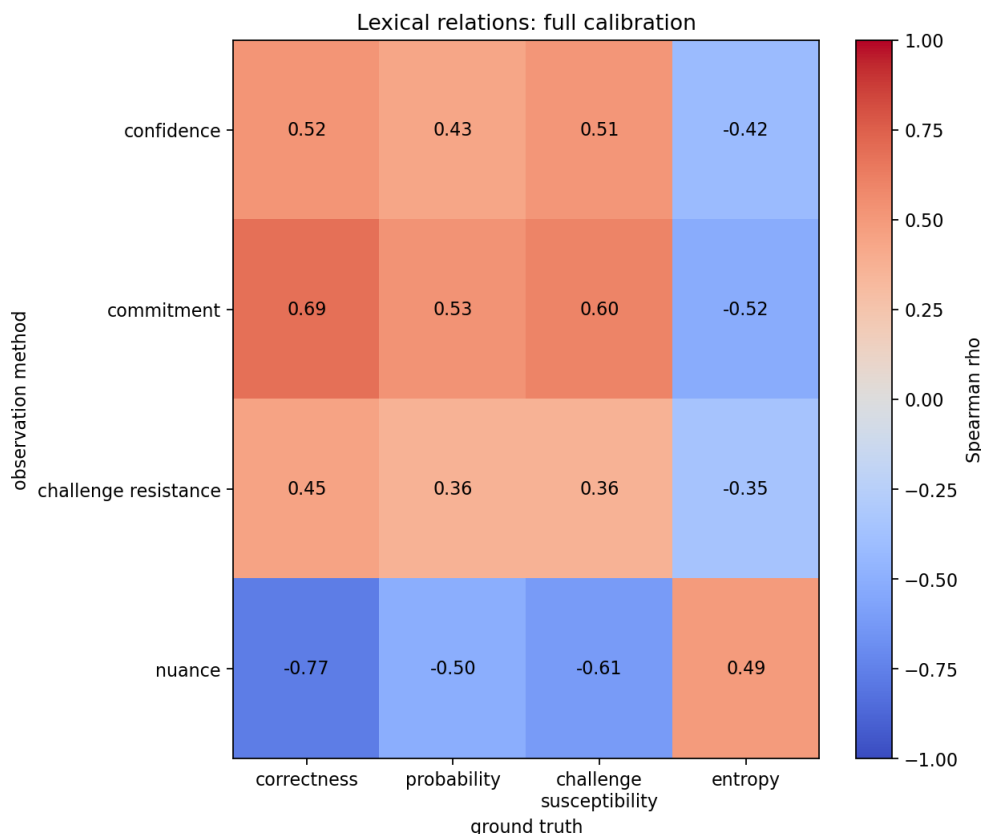


Figure 6.3: Full calibration matrix for lexical relations.

method on Synonyms.

Figure 6.4 uses the same four-by-four layout for TriviaQA. The challenge-resistance report and challenge-susceptibility ground truth have an association of -0.162, with a 95% interval of [-0.294, -0.009]. The negative sign is the expected sign for these oppositely oriented scales, unlike the positive Synonyms result in Fig. 6.3. Its small magnitude supports only limited monotonic alignment and does not by itself demonstrate discriminant validity. Confidence, commitment, and nuance recover their expected matched directions, but their relative profiles differ from those in the lexical-relation data.

Figure 6.5 shows the correlation between self-reported variety and computed list variance, which is 0.224 and has a 95% confidence interval of [0.083, 0.365]. The rows and columns include only those variables that have been defined for list generation. Variety is also associated with the temperature control at 0.244 [0.109, 0.370]. The similar magnitudes limit a construct-specific interpretation of the variety report.

Figure 6.6 places the temperature control on both axes for diagnostic purposes. Its diagonal correlation of 1.00 follows mechanically from comparing the variable with itself. Variety and temperature correlate at 0.244, with a 95% interval of [0.109, 0.370], while temperature and list variance correlate at 0.183 [0.034, 0.317]. Both weak positive associations have intervals excluding zero. Temperature is therefore related to the generated lists and their reported variety in this sample, despite not being shown directly during observation.

The pooled correlation also conceals differences among keywords. Across the 20 temper-

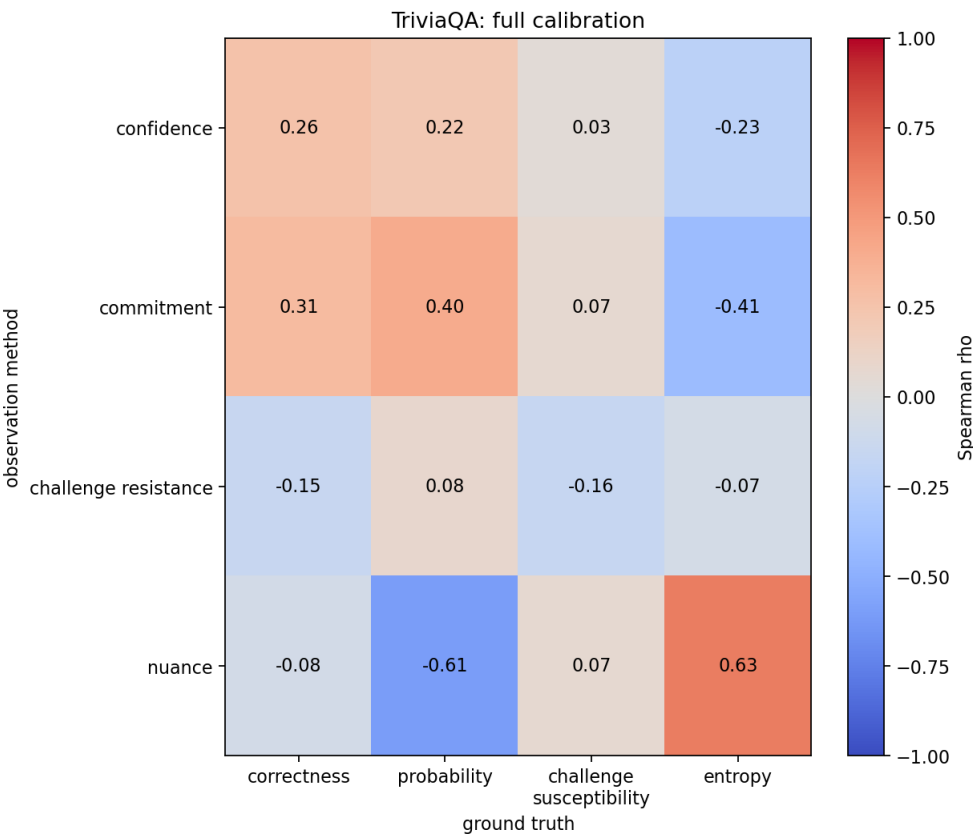


Figure 6.4: Full calibration matrix for TriviaQA.

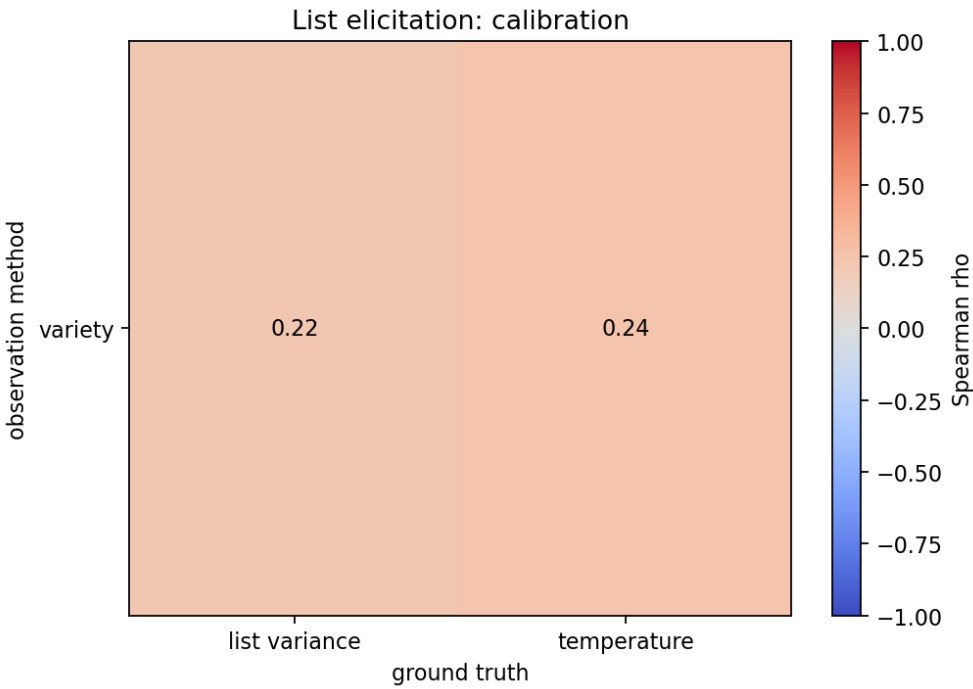


Figure 6.5: List-elicitation calibration matrix.

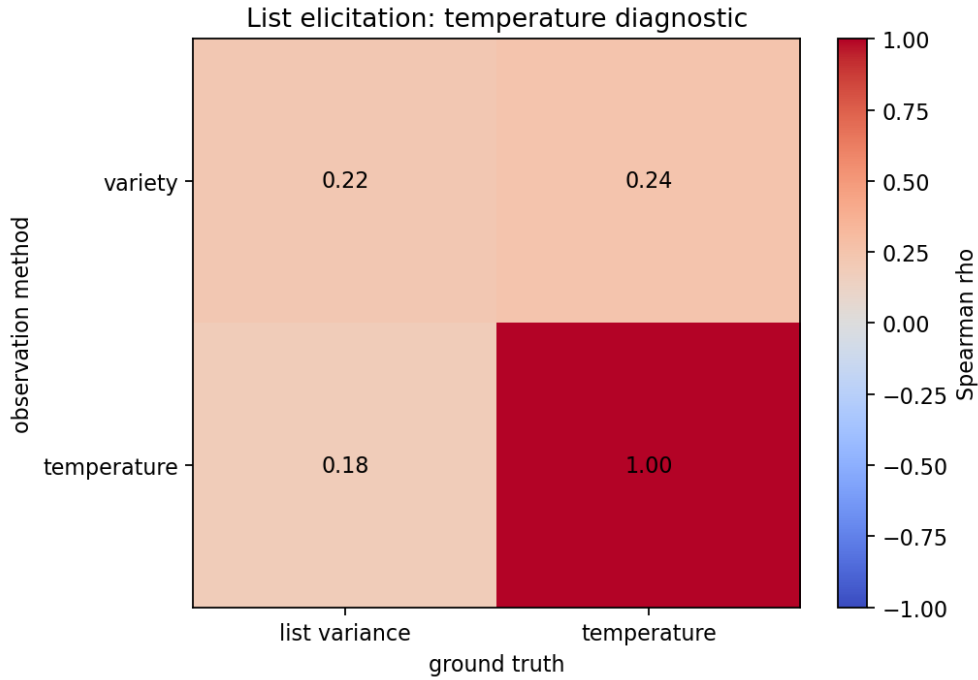


Figure 6.6: Temperature-control calibration matrix for list elicitation.

atures available for each of 10 keywords, the within-keyword correlations range from -0.488 to 0.756; seven are positive, with a mean of 0.192 and a median of 0.319 (Table 6.3). These small, non-independent comparisons are exploratory and are not adjusted for multiplicity. They show that the temperature–variance relationship is heterogeneous rather than uniform across prompts.

Table 6.3: Per-keyword temperature diagnostic.

Per-keyword temperature diagnostic	Value
Keywords / temperatures per keyword	10 / 20
Positive correlations	7 / 10
Mean within-keyword rho	0.192
Median within-keyword rho	0.319
Range of within-keyword rho	[-0.488, 0.756]

In Fig. 6.7 the impurity report is compared with the Gini impurity of the induced yes/no candidate partition. Across 200 games, the correlation is 0.306 with a 95% interval of [0.171, 0.437]. This is a modest calibration result; the separate steering sample did not provide a nondegenerate high–low contrast.

Together, Figs. 6.3 and 6.4 cover the lexical-relation and TriviaQA pools; the remaining figures address list variety, temperature, and twenty-questions impurity. Thus, calibration covers all four dataset families. These correlations address measurement validity but do not answer the causal generalization question in RQ3.

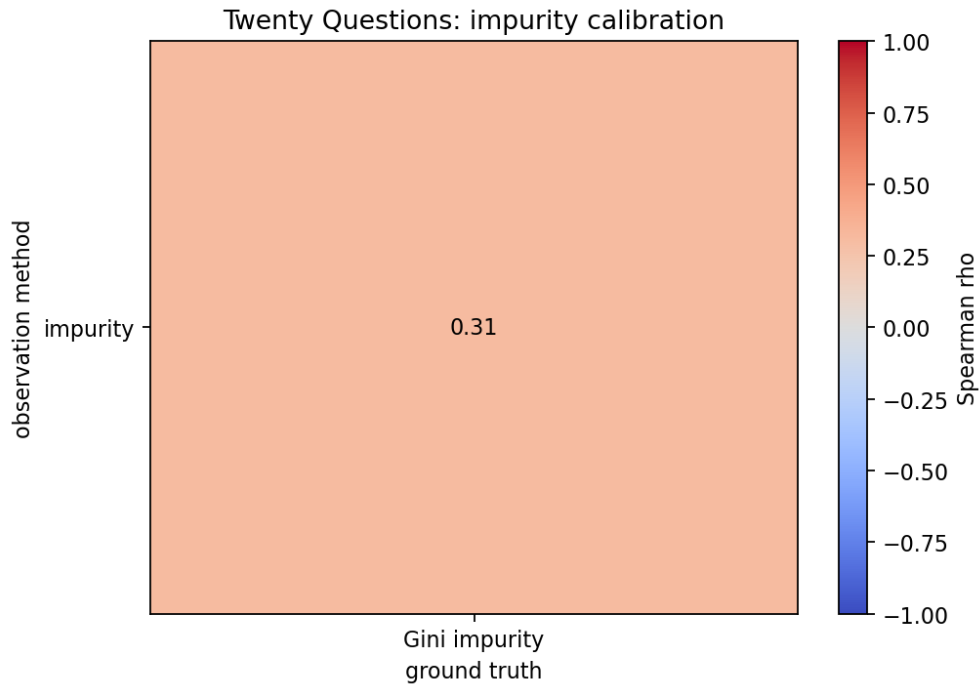


Figure 6.7: Twenty-questions impurity calibration result.

6.3 Steering results and generalization

The steering procedure supports position-controlled, bidirectional sweeps for confidence, commitment, nuance, variety, and impurity, but completed results are limited to TriviaQA confidence and Synonyms commitment. Each analysis uses 30 extraction trials, 60 disjoint candidate trials, and a balanced 30-trial test set. TriviaQA draws from 200 prepared seeds; Synonyms has 201 prepared pairs, with 67 per relation tier. Table 6.4 distinguishes these samples from the calibration data.

Table 6.4: Completed steering analyses.

Steering run	Source pool	Extraction / candidates / selected test	Layers and scale	Inference
TriviaQA confidence	200 seeds	30 / 60 / 30 trials	Layers 0, 2, ..., 26, 27; both signs; $\alpha = 5.0$	Little consistent PANL separation; stronger CC readout effect
Synonyms commitment	201 pairs	30 / 60 / 30 trials	Layers 0, 2, ..., 26, 27; both signs; $\alpha = 5.0$	Strong late-layer CC readout effect; no comparable PANL separation

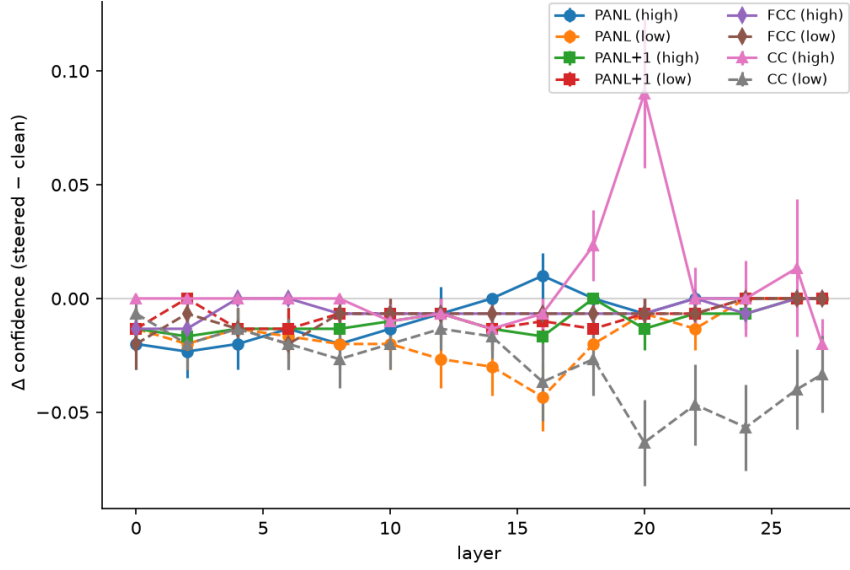


Figure 6.8: TriviaQA confidence steering by layer and position on the balanced 30-trial test sample.

For each layer ℓ , position p , and direction α , the plotted midpoint change and its uncertainty are aggregated over the test set as

$$\begin{aligned}\overline{\Delta m}(\ell, p, \alpha) &= \frac{1}{N} \sum_{i=1}^N \Delta m(\theta_i; \ell, p, \alpha), \\ \text{SEM}(\ell, p, \alpha) &= \text{sem}\left(\{\Delta m(\theta_i; \ell, p, \alpha)\}_{i=1}^N\right), \\ \bar{d}(\ell, p, \alpha) &= \frac{1}{N} \sum_{i=1}^N d_i(\ell, p, \alpha).\end{aligned}\tag{8}$$

Figure 6.8 reports the mean midpoint change and SEM defined by Equation (8), relative to the clean condition, over the selected 30 test trials. PANL, PANL+1, and FCC show little consistent separation between the two steering directions. The clearer response appears at CC in the later layers, where the directions separate on opposite sides of the clean condition. This is a descriptive pattern from the layer sweep; the SEM bars quantify within-sample variation but do not supply confirmatory tests across the inspected layers.

Figure 6.9 uses the same 30-trial design for commitment. The two steering directions produce a clear bidirectional response at CC in the later layers, while PANL, PANL+1, and FCC remain close to zero. No comparable answer-adjacent effect appears. The SEM bars describe variation within the selected test trials but do not account for layer selection, extraction-pool variation, or differences between datasets.

In the impurity analysis, all 40 twenty-questions candidates received the same self-reported midpoint, 0.65. With no high-low contrast, the selection rule stopped before steering; consequently, no impurity effect, peak layer, or bidirectionality estimate is reported.

The constant sample may reflect an invariant response induced by the prompt, insufficient variation in the candidate pool, or poor discrimination among trials. It supplies no contrast with which to test a causal effect.

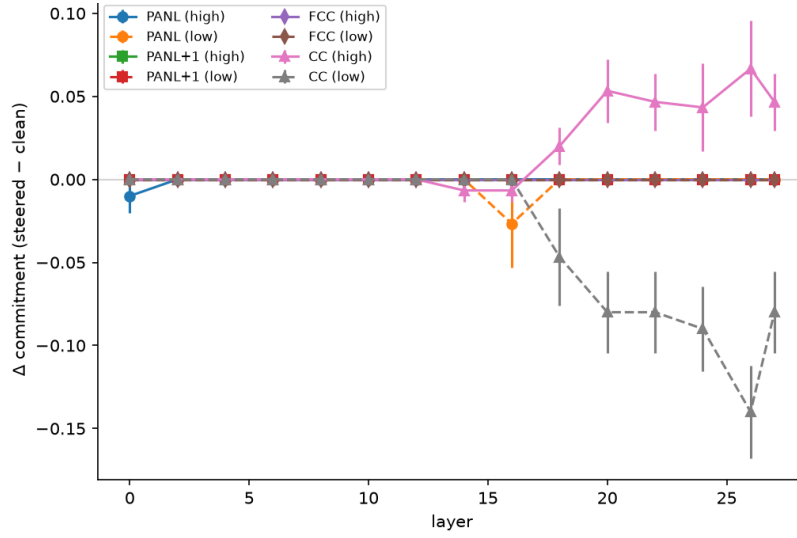


Figure 6.9: Synonyms commitment steering by layer and position on the balanced 30-trial test sample.

Across TriviaQA and Synonyms, the clearest response occurs under direct intervention at the final readout cue, while neither completed analysis shows a consistent answer-adjacent pattern. The results establish that the two reports can be manipulated at their readout cue. The answer-adjacent dissociation specified by RQ2 is absent from the reduced run, and the shared pattern proposed in RQ3 is absent from the two completed analyses. Generalization to the remaining constructs lies outside the available causal evidence. For RQ4, the evidence is compatible with final-cue computation and does not favour cache-and-retrieval.

Chapter 7

Discussion: The Ontological Argument

The calibration matrix replaces the broad question “does the model have a belief?” with a measurable one: which answer-related properties covary with each prompted report? The labels alone cannot answer it. Confidence correlates with correctness and answer-sequence probability in both datasets; the point estimate is higher for correctness (0.516 versus 0.433 for Synonyms and 0.262 versus 0.216 for TriviaQA). Because the analysis does not test the difference between these dependent correlations, this ordering is descriptive. It nevertheless gives the confidence–correctness hypothesis support in both samples while showing that confidence also follows the model’s support for its answer. Benzon’s account of categories as relational structures (Benzon, 1987) motivates this focus on the report’s empirical relations rather than its name.

Commitment shows a different ordering across datasets. In Synonyms its correlation is higher with correctness than with probability (0.687 versus 0.530); in TriviaQA the corresponding estimates are 0.306 and 0.405. This reversal supports an association with probability but not a task-independent or construct-specific interpretation. Nuance also differs across datasets. In Synonyms it is more strongly associated with correctness in the negative direction than with entropy (-0.769 versus 0.489), whereas in TriviaQA it is weakly associated with correctness and more strongly associated with probability and entropy (-0.609 and 0.627). The constrained binary lexical task therefore produces a profile organized largely by correctness, while the open-ended factual task more clearly separates sequence support and entropy. The three reports resist reduction to a single “belief score.” On the relational interpretation, the object of study is the association among the prompt, answer, readout, and comparator.

Challenge provides the clearest failure of a common interpretation. Reported resistance should be negatively associated with computed susceptibility. TriviaQA has the expected but small negative association (-0.162), whereas Synonyms has a positive association (0.360). The challenge hypothesis is therefore unsupported for lexical relations and receives only limited support for TriviaQA. List variety is weakly associated with list variance (0.224), but its

similar association with hidden temperature (0.244) prevents a construct-specific conclusion. The temperature–variance relationship also varies considerably by keyword. These results show that prompt and task structure affect what a report tracks.

Benzon’s later work supplies the vocabulary for why dataset structure matters to this relational account: concrete/abstract categories and the twenty-questions game as probes of conceptual structure (Benzon, 2023b), and a middle descriptive level of nested probability distributions between surface behaviour and implementation (Benzon, 2023a). Neither proposal is used here as a validated model of transformers.

The lexical and twenty-questions results extend this relational account. Benzon’s inheritance structure treats strict synonymy as a conceptual boundary rather than generic similarity (Benzon, 1987); the commitment correlation may therefore reflect sensitivity to that boundary. A twenty-questions query partitions the candidate set (Benzon, 2023b), and the impurity report has a modest positive correlation with the partition’s Gini impurity (0.306, 95% CI [0.171, 0.437]). More balanced splits tend to receive higher reports, while the constant steering sample leaves the mechanism unresolved. The result concerns measurement across tasks rather than a shared causal mechanism.

Benzon’s studies of recurring narrative patterns under varied prompts (Benzon, 2024) illustrate how a distributional regularity may be useful before it is tied to a specific semantic object. The calibration matrices reveal regularities in prompted reports but cannot determine whether they arise from a single vector, a hierarchy of distributions, or a prompt-specific retrieval strategy. The confidence and commitment interventions show that each final report can be changed at its readout cue in one model and prompt setting. The absence of a consistent answer-adjacent effect leaves persistence across prompt positions unsupported. Direct CC sensitivity is compatible with a report assembled at the final cue, while the task-dependent calibration profiles support a relational, measurement-dependent interpretation. The observer-relative reading is therefore the narrowest account supported here; representational-state and linguistic-performance readings remain possible.

Measurement choices help define the empirical object. Confidence has an obvious comparison with correctness, whereas commitment, nuance, challenge, variety, and impurity depend on a chosen ground truth, such as answer-sequence probability or answer entropy. Observation method and ground truth therefore specify the relation under study, consistent with Benzon’s treatment of synonym boundaries and Twenty Questions (Benzon, 1987, 2023b). Forced cues determine the readout position, tokenizer behaviour determines whether PANL can be isolated, and the sampling rule determines whether high and low report groups can be formed. The constant impurity reports demonstrate how the measurement procedure can prevent a causal test.

Chapter 8

Limitations

The main empirical limitation is scope. The reproduction uses Qwen2.5-7B-Instruct and the reported TriviaQA conditions. The findings therefore do not establish generality across architectures, datasets, or reasoning models.

The behavioural pool contains 1,000 trials, but some interventions use only dozens of held-out observations. Standard errors in the steering figures quantify variation within those samples; they do not account for layer selection or multiple comparisons. The steering curves therefore support qualitative pattern comparisons rather than inference about selected extrema. The 200-trial calibration matrices are likewise exploratory. Bootstrap intervals are reported for selected comparisons but are not adjusted for the full matrix or for the per-keyword list diagnostics.

The causal generalization evidence consists of confidence on TriviaQA and commitment on Synonyms. The twenty-questions sample yielded a constant impurity report and therefore no steering contrast. This scope is insufficient to describe cache-and-retrieval as a construct-general architecture.

The ground truths are operational measures with distinct limitations. Alias matching measures overlap with accepted answers rather than semantic correctness and can reject valid paraphrases, particularly for open-ended TriviaQA answers. The forced Yes/No lexical task is less exposed to that limitation but also differs structurally from TriviaQA. Answer-sequence probability varies with answer length, and answer entropy depends on sampling and tokenization. Challenge susceptibility relies on generated counter-evidence, while impurity is computed from the recorded candidate universe rather than the latent set remaining at each turn. List variance and its relationship with temperature vary across keywords. Each measure addresses a stated empirical question, but none directly measures belief.

Finally, the ontological argument is an interpretation of the measurement design and causal evidence, not a metaphysical resolution. Its conclusions remain conditional on the prompts and readout procedures used here.

Chapter 9

Conclusion and Future Work

This thesis examined how an LLM produces self-reports about its answers. In the reduced Qwen experiment, the clearest confidence change followed direct intervention at the final report cue, while answer-adjacent positions showed little consistent separation. Within the limits of a single model and a reduced protocol, this result establishes direct manipulability of the readout but does not favour cached retrieval over computation at the final cue.

vonto separates the self-report observation method from the computed ground truth, allowing each report to be compared with a task-appropriate criterion rather than treating correctness as universal. Calibration covers four dataset families, but the associations depend on the task. The lexical-relation reports are strongly organized by binary correctness, while commitment and nuance in TriviaQA align more clearly with answer-sequence probability and answer entropy. Variety and impurity show modest positive associations with their designated measures, although the variety result is not specific relative to the temperature control. The challenge results do not support a common relationship across the two datasets. The causal evidence does not yet support a shared architecture across constructs, and the constant impurity reports expose a limitation of the elicitation design. Taken together, these findings support an observer-relative account of the measured reports while leaving representational-state and linguistic-performance interpretations open.

Future work should complete the position-controlled steering suite with larger, preregistered trial pools and uncertainty intervals for every aggregate effect. It should also evaluate the AC regeneration control, compare steering magnitudes, and extend patching, noising, representation swaps, probing, and attention interventions to the generalized constructs. Tests on Gemma and additional model families are required before drawing architectural conclusions. Finally, adversarial prompt variants and alternative ground truths could reveal which properties of a belief report remain stable when the extraction rule changes.

References

- Benzon, W. L. (1987). Ontology in knowledge representation. <https://doi.org/10.2139/ssrn.1508554>
- Benzon, W. L. (2023a). *Chatgpt tells stories, and a note about reverse engineering* [Unpublished working paper].
- Benzon, W. L. (2023b). Chatgpt’s ontological landscape [Working paper]. <https://doi.org/10.2139/ssrn.4620133>
- Benzon, W. L. (2024). *Chatgpt tells 20 versions of its prototypical story, with a short note on method* [Unpublished working paper].
- Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630, 625–630. <https://doi.org/10.1038/s41586-024-07421-0>
- Haukoos, J. S., & Lewis, R. J. (2005). Advanced statistics: Bootstrapping confidence intervals for statistics with difficult distributions. *Academic Emergency Medicine*, 12(4), 360–365. <https://doi.org/10.1197/j.aem.2004.11.018>
- Joshi, M., Choi, E., Weld, D. S., & Zettlemoyer, L. (2017). TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, 1601–1611. <https://aclanthology.org/P17-1147/>
- Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Dodds, Z. H., DasSarma, N., Tran-Johnson, E., Johnston, S., El-Showk, S., Jones, A., Elhage, N., Hume, T., Chen, A., Bai, Y., Bowman, S., Fort, S., . . . Kaplan, J. (2022). Language models (mostly) know what they know. <https://arxiv.org/abs/2207.05221>
- Kumaran, D., Conmy, A., Barbero, F., Osindero, S., Patraucean, V., & Veličković, P. (2026). How do LLMs compute verbal confidence? <https://arxiv.org/abs/2603.17839>
- Nerantzaki, K., Efklides, A., & Metallidou, P. (2021). Epistemic emotions: Cognitive underpinnings and relations with metacognitive feelings. *New Ideas in Psychology*, 63, Article 100904. <https://doi.org/10.1016/j.newideapsych.2021.100904>
- Qwen Team. (2024). *Qwen2.5 technical report* (tech. rep.). <https://arxiv.org/abs/2412.15115>
- Santus, E., Gladkova, A., Evert, S., & Lenci, A. (2016). The CogALex-V shared task on the corpus-based identification of semantic relations. *Proceedings of the 5th Workshop on Cognitive Aspects of the Lexicon*, 69–79. <https://aclanthology.org/W16-5309/>

- Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., & Hooi, B. (2023). Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. <https://arxiv.org/abs/2306.13063>